A Course End Project

Smart Irrigation System

IOT LABORATORY (A8603)

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

This is to certify that the Course End Project titled **“Smart Irrigation System”** is carried out by **B Swathi(24881A0576), G Spoorthi Priya (24881A0588)**, towards **A8603 – IOT Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology in Computer Science and Engineering** during the Academic Year 2025–26.

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Check List

S. No.	Content	Status (✓ / ✗)
1	Abstract	✓
2	Objectives	✓
3	Problem Statement	✓
4	Algorithm	✓
5	Flow Chart	✓
6	Source Code	✓
7	Input	✓
8	Output	✓
9	Conclusion	✓
10	Future Scope	✓
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Course End Project

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Table of Content

Chapter 1

Abstract

This project titled **Smart Irrigation System** with the advancement of technology, the application of computer science in agriculture has led to the development of intelligent and automated systems. One such innovation is the Smart Irrigation System, which aims to optimize water usage and improve agricultural efficiency through automation and data-driven decision-making.

The proposed system integrates embedded systems, sensor networks, and Internet of Things (IoT) technologies. Soil moisture sensors are used to collect real-time data from the field, which is processed by a microcontroller such as Arduino or NodeMCU. Based on the analyzed data and predefined threshold values, the system automatically controls the irrigation process by activating or deactivating the water pump through a relay module.

The system also incorporates temperature and humidity sensors to enhance environmental monitoring. Using IoT connectivity, the collected data is transmitted to a cloud platform, where it can be visualized through a web or mobile application. This enables remote monitoring, data logging, and intelligent control of irrigation, reducing the need for manual intervention.

From a computer science perspective, the project involves programming, data acquisition, real-time processing, and communication between hardware and software components. It demonstrates the practical implementation of embedded systems, wireless communication, and automation techniques.

The Smart Irrigation System is a cost-effective, scalable, and efficient solution that conserves water, reduces labor, and increases crop yield. It highlights the role of computer science in solving real-world problems and promoting sustainable agriculture.

Keywords: Smart Irrigation System, Internet of Things (IoT), Soil Moisture Sensor, Arduino, Automation, Embedded Systems, Water Management, Wireless Communication, Sensor Networks, Precision Agriculture, Data Monitoring, Cloud Computing, Remote Control

Chapter 2

Problem Statement

Efficient water management is a major challenge in modern agriculture. Traditional irrigation methods, such as manual watering or timer-based systems, often lead to over-irrigation or under-irrigation, resulting in water wastage, increased labor costs, and reduced crop yield. Farmers lack a system that can dynamically respond to actual soil and environmental conditions, making irrigation inefficient and resource-intensive.

The problem is further compounded in large-scale farming, where monitoring multiple fields manually is time-consuming and prone to errors. Additionally, many existing irrigation systems do not provide real-time data or remote monitoring capabilities, limiting the ability of farmers to make timely decisions. Therefore, there is a need for a smart, automated irrigation system that can:

- Monitor soil moisture, temperature, and humidity in real-time,
- Automatically control irrigation based on data,
- Enable remote monitoring and management through IoT,
- Optimize water usage to conserve resources,and
- Improve crop productivity and reduce labor effort

By implementing this system, farmers can conserve water, reduce labor, increase productivity, and adopt sustainable agricultural practices. The Smart Irrigation System demonstrates how embedded systems, sensor networks, and IoT technologies can be applied to real-world problems in agriculture, providing a practical and scalable solution for modern farming.

Chapter 3

Objectives

1. Automated Irrigation Control: To automatically monitor soil moisture levels and control the water pump to irrigate crops only when required, reducing human intervention.

2. Real-Time Environmental Monitoring: To measure and track environmental parameters such as soil moisture, temperature, and humidity for better irrigation decisions.

3. Remote Monitoring and IoT Integration: To provide remote access via a mobile application or web interface, allowing farmers to monitor and control irrigation from anywhere.

4. Water Conservation and Efficiency: To optimize water usage by delivering the right amount of water at the right time, preventing over-irrigation and wastage.

5. Data Logging and Analysis: To record historical sensor data for analysis, helping farmers make data-driven decisions and improve crop yield over time.

6. Practical Application of Embedded Systems and IoT: To demonstrate how sensors, microcontrollers, and IoT technologies can be integrated into a real-world precision agriculture system.

Chapter 4

Algorithm

1. **System Initialization:**
   (a) Power ON the system.
   (b) Initialize microcontroller (Arduino/NodeMCU).
   (c) Initialize sensors: Soil Moisture, Temperature, Humidity.
   (d) Establish IoT connectivity (optional).

2. **Read Sensor Data:**
   (a) Continuously read soil moisture value.
   (b) Read temperature and humidity.

3. **Compare Threshold Values:**
   (a) Define threshold for soil moisture.
   (b) If soil moisture ¡ threshold → irrigation required.
   (c) If soil moisture threshold → no irrigation.

4. **Control Water Pump:**
   (a) Turn ON pump if irrigation required.
   (b) Continue until threshold reached.
   (c) Turn OFF pump once threshold is met.

5. **IoT Integration (Optional):**
   (a) Send real-time data to cloud.
   (b) Display on mobile/web app.

(c) Allow manual remote control.

6. **Data Logging (Optional):**

    (a) Store sensor readings.

    (b) Analyze historical data for optimization.

7. **Continuous Monitoring:**

    (a) Repeat steps 2–6 for real-time irrigation.

8. **End:**

    (a) System runs until manually turned off.

Chapter 5

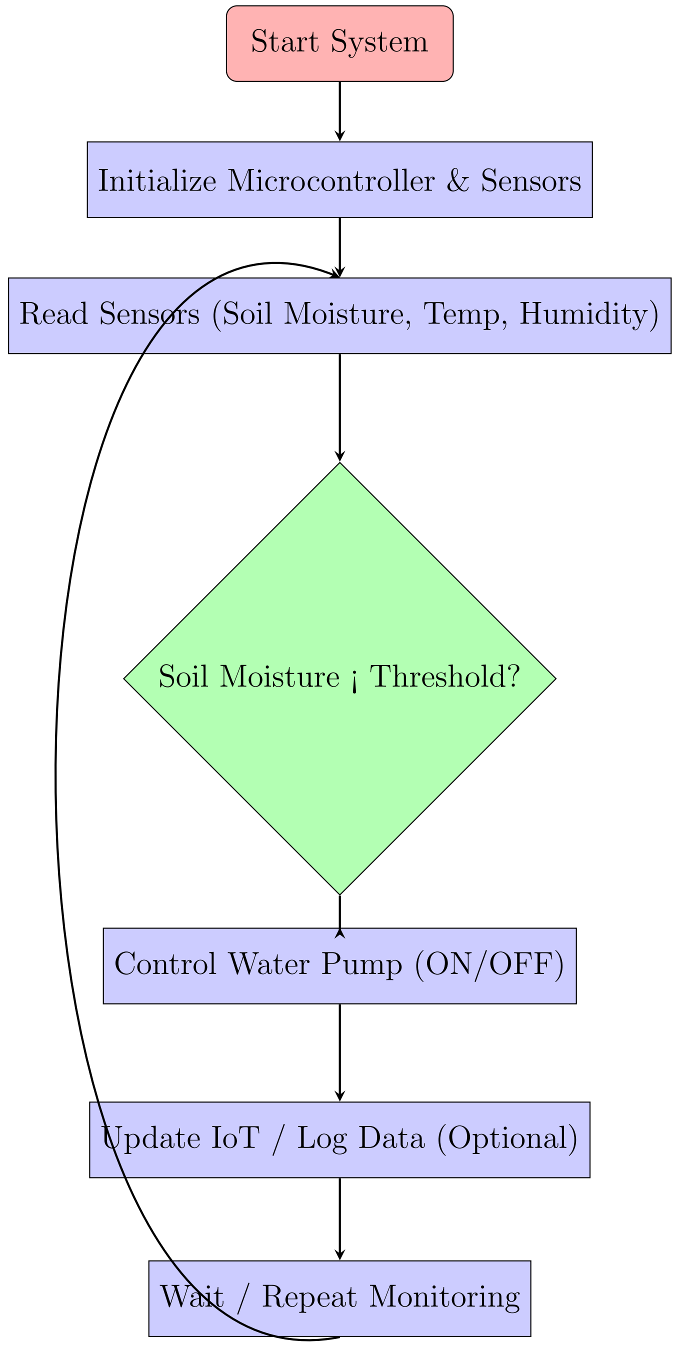
Proposed System

The proposed Smart Irrigation System is designed to automate irrigation and optimize water usage in agriculture. It continuously monitors soil and environmental conditions using soil moisture sensors, temperature sensors, and humidity sensors. The collected data is processed by a microcontroller (Arduino/NodeMCU), which makes real-time decisions to control the water pump via a relay module. By activating irrigation only when necessary, the system prevents overwatering and under-watering, ensuring crops receive the optimal amount of water while reducing labor and resource wastage.

To enhance usability and modern functionality, the system incorporates IoT technology, allowing farmers to monitor and control irrigation remotely through mobile or web applications. Sensor data can be transmitted to the cloud for real-time visualization, historical analysis, and alert notifications. This enables farmers to make data-driven decisions, adjust irrigation schedules based on actual field conditions, and improve overall water management efficiency.

The system is modular, scalable, and adaptable for different farm sizes and crop types. Its design promotes sustainable agriculture by conserving water, reducing labor costs, and improving crop productivity. By combining automation, sensor networks, and IoT connectivity, the proposed Smart Irrigation System provides a modern, intelligent, and efficient solution for precision farming, making it a practical application of computer science and embedded systems in agriculture.

5.1 Flow Chart



5.2 Working Model

- **Objective:** To automate irrigation based on real-time soil moisture, temperature, and humidity readings, conserving water, reducing manual labor, and improving crop productivity.

- **Components Required:**
  - Microcontroller: Arduino UNO or NodeMCU (for IoT)
  - Sensors:
    * Soil Moisture Sensor (detects water content in soil)
    * DHT11 or DHT22 Sensor (measures temperature & humidity)
  - Actuator: Water Pump (controlled via Relay Module)
  - Power Supply: 5V or 12V depending on pump and microcontroller
  - IoT Module (Optional): ESP8266 or ESP32 for remote monitoring
  - Relay Module: To switch the water pump ON/OFF safely
  - Connecting wires, breadboard or PCB

- **System Workflow:**
  - Initialization: Microcontroller powers up, initializes sensors and relay module.
  - Data Collection: Sensors measure soil moisture, temperature, and humidity.
  - Decision Making:
    * If soil moisture < predefined threshold → Pump turns ON.
    * Else → Pump stays OFF.
  - Water Pump Control: Relay module switches the water pump accordingly.
  - IoT Monitoring (Optional): Sensor readings are sent to a cloud platform (e.g., Blynk, Thingspeak) for real-time monitoring.
  - Data Logging (Optional): System logs the data for future analysis and irrigation optimization.
  - Continuous Operation: Steps repeat at a fixed interval (e.g., every 10 minutes).

- **Implementation Steps:**
  - Connect sensors and relay module to the microcontroller.

  - Program the microcontroller: read sensors → check thresholds → control pump → log/send data.
  - Test the system with simulated soil moisture levels to verify pump control.
  - (Optional) Connect to IoT platform for remote monitoring and alerts.

- **Advantages of This Working Model:**
  - Fully automated irrigation system.
  - Water-efficient: irrigates only when necessary.
  - Scalable and modular: additional sensors or pumps can be added.
  - Remote monitoring via IoT.
  - Easy to demonstrate in a project presentation.

Chapter 6

Source Code

```
#include <DHT.h>

// Pin Definitions
#define SOIL_SENSOR_PIN A0
#define RELAY_PIN 8
#define DHT_PIN 2
#define DHT_TYPE DHT11

// Constants
#define SOIL_THRESHOLD 500
#define DELAY_INTERVAL 600000 // 10 minutes

DHT dht(DHT_PIN, DHT_TYPE);

void setup() {
  Serial.begin(9600);
  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, LOW); // Pump OFF at start
  dht.begin();
  Serial.println("Smart Irrigation System Initialized");
}

void loop() {
  // Read Soil Moisture
  int soilMoistureValue = analogRead(SOIL_SENSOR_PIN);
  Serial.print("Soil Moisture: ");
  Serial.println(soilMoistureValue);

```

```
  // Read Temperature & Humidity
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();

  if (isnan(temperature) || isnan(humidity)) {
    Serial.println("Failed to read from DHT sensor!");
  } else {
    Serial.print("Temperature: "); Serial.print(temperature); Serial.println("  C ");
    Serial.print("Humidity: "); Serial.print(humidity); Serial.println(" %");
  }

  // Decision Making
  if (soilMoistureValue < SOIL_THRESHOLD) {
    digitalWrite(RELAY_PIN, HIGH); // Pump ON
    Serial.println("Pump ON - Soil is dry");
  } else {
    digitalWrite(RELAY_PIN, LOW);  // Pump OFF
    Serial.println("Pump OFF - Soil is wet");
  }

  // Optional: IoT upload here (Blynk, ThingSpeak)

  Serial.println("---------------------------");
  delay(DELAY_INTERVAL);
}
```

Listing 6.1: Arduino Source Code for Smart Irrigation System

Chapter 7

Input

The Smart Irrigation System takes the following inputs to operate efficiently:

- **Soil Moisture Sensor Reading:** Measures the water content in the soil. This value determines whether irrigation is needed.
- **Temperature:** Measured using a DHT11 or DHT22 sensor to monitor ambient conditions and adjust irrigation scheduling if necessary.
- **Humidity:** Also measured using the DHT sensor to provide environmental context for irrigation.
- **User-Defined Threshold (Optional):** The soil moisture threshold value set by the user, which decides when the water pump should turn ON or OFF.
- **Timing Interval (Optional):** The frequency at which the system checks sensor readings and updates pump control.

These inputs allow the system to make decisions automatically, ensuring water-efficient and timely irrigation of crops.

Chapter 8

Output

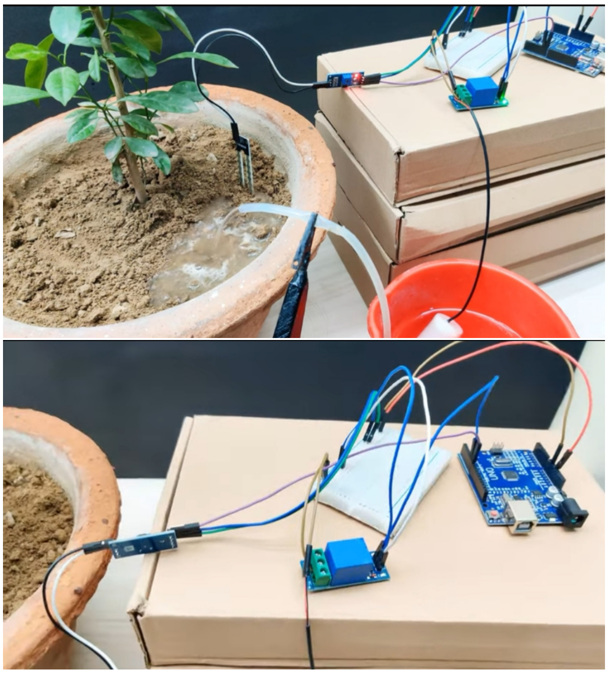


Figure 8.1: Smart Irrigation System

Chapter 9

Conclusion

The Smart Irrigation System effectively demonstrates the potential of automation in modern agriculture. By using real-time soil moisture, temperature, and humidity readings, the system ensures that crops receive the right amount of water at the right time. This targeted approach conserves water, reduces wastage, and lowers manual labor, which is especially important in regions facing water scarcity. The integration of sensors with a microcontroller allows the system to make intelligent decisions without constant human supervision.

One of the key advantages of this system is its ability to maintain optimal soil conditions for crop growth. By continuously monitoring environmental parameters, the system can respond immediately to changes in soil moisture or climate, preventing over-irrigation or under-irrigation. This helps improve crop yield and quality while minimizing the risk of plant stress caused by irregular watering schedules.

The inclusion of IoT capabilities further enhances the functionality of the system. With cloud connectivity, users can monitor soil and environmental conditions remotely, receive alerts, and even control irrigation schedules via a mobile application or web interface. This feature makes the system scalable and adaptable to both small farms and large agricultural operations, providing farmers with valuable insights into their fields in real-time.

Overall, the Smart Irrigation System is a practical, efficient, and sustainable solution for modern agriculture. It combines technology, efficiency, and environmental consciousness to improve crop management practices. This project not only emphasizes the importance of automation in farming but also provides a foundation for future enhancements, such as AI-based decision-making, predictive irrigation, and integration with larger smart farm systems.

Chapter 10

Future Scope

The Smart Irrigation System offers multiple avenues for enhancement and can be developed further to make it more efficient, sustainable, and intelligent:

- **Integration with AI and Machine Learning:** Use historical and real-time sensor data to predict irrigation requirements, optimize water usage, and adapt to changing weather patterns.
- **Solar-Powered Operation:** Implement solar panels and energy storage to make the system self-sufficient and suitable for off-grid farming areas.
- **IoT Integration and Remote Access:** Enable cloud connectivity for real-time monitoring, data logging, notifications, and remote control via mobile or web applications.
- **Automated Fertilization and Pest Control:** Extend the system to manage fertilization and pest control based on sensor data, creating a full smart farm solution.
- **Alerts and Predictive Maintenance:** Introduce notifications for maintenance of pumps, sensors, and irrigation equipment to reduce downtime.
- **Scalability and Modular Design:** Design the system to easily add more sensors, pumps, or irrigation zones for larger agricultural fields.
- **Weather Forecast Integration:** Use weather API data to schedule irrigation based on rainfall prediction and avoid unnecessary water usage.
- **Data Analytics and Reporting:** Maintain historical data and generate reports for farm management, productivity tracking, and decision-making.

Chapter 11

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